

NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

LECTURE 4
Output, the Interest Rate, and the Exchange Rate

READINGS

1. Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapter 19.

INTRODUCTION

1. In some countries, exchange rate is not a policy instrument. Rather, it is determined in the foreign exchange market.
 - (a) What determines the exchange rate?
 - (b) How can policymakers affect it?
2. To answer these questions, we reintroduce financial markets and examine the implications of equilibrium in both the goods market and financial markets, including the foreign exchange market.
3. This allows us to characterize the joint movements of output, the interest rate, and the exchange rate in an open economy.
4. This is an extension of the IS-LM model and is known as the **Mundell-Fleming model**.

EQUILIBRIUM IN THE GOODS MARKET

1. Recall that equilibrium in the goods market gives:

$$Y = C(Y - T) + I(Y, r) + G + X(Y^*, \epsilon) - \frac{IM(Y, \epsilon)}{\epsilon} \quad (1)$$

2. For the goods market to be in equilibrium, output (LHS) must be equal to the demand for domestic goods (RHS).
3. Since

$$NX(Y, Y^*, \epsilon) = X(Y^*, \epsilon) - \frac{IM(Y, \epsilon)}{\epsilon} \quad (2)$$

then,

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \epsilon) \quad (3)$$

As $\epsilon = EP/P^*$, and we let $P/P^* = 1$, then, $\epsilon = E$. Therefore, we have at goods market equilibrium:

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, E) \quad (4)$$

EQUILIBRIUM IN FINANCIAL MARKETS

1. When we looked at financial markets in the IS-LM model, we assumed that people choose between two financial assets: money and bonds.
2. Now that we look at a financially open economy, we must take into consideration that they can choose between domestic bonds and foreign bonds.

Domestic Bonds vs Foreign Bonds

1. As we look at the choice between domestic bonds and foreign bonds, we assume that financial investors, domestic or foreign, go for the highest expected rate of return.
2. This implies that, in equilibrium, both domestic bonds and foreign bonds must have the same expected rate of return; otherwise, investors would be willing to hold only one or the other, but not both, and this could not be an equilibrium.
3. This assumption implies that the following arbitrage relation – the interest parity condition – must hold:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (5)$$

where i_t is the domestic interest rate, i_t^* is the foreign interest rate, E_t is the current exchange rate, and E_{t+1}^e is the expected future exchange rate.

4. Reorganize this gives:

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (6)$$

5. When E_{t+1}^e is taken as given, then we have:

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e \quad (7)$$

6. This relation tells us that current exchange rate depends on the domestic interest rate, the foreign interest rate, and the expected future exchange rate.
 - (a) An increase in the domestic interest rate leads to an increase in the exchange rate.
 - (b) An increase in the domestic interest rate leads to a decrease in the exchange rate.

- (c) An increase in the expected future exchange rate leads to an increase in the current exchange rate.

7. Figure 1 shows the relation between the interest rate and the exchange rate implied by interest parity.

[Figure 1: Interest parity relation]

8. The relation is drawn for a given expected future exchange rate, $\bar{E}^e$, and a given foreign interest rate, i^* , and is represented by an upward-sloping line: the higher the domestic interest rate, the higher the exchange rate.
9. When the domestic interest rate is equal to the foreign interest rate ($i = i^*$), the exchange rate is equal to the expected future exchange rate ($E = \bar{E}^e$).
10. This implies that the line corresponding to the interest parity condition goes through point 0 in Figure 1.
11. This relation suggests that an increase in domestic interest rate relative to foreign interest rate leads to an appreciation of domestic currency.

PUTTING GOODS AND FINANCIAL MARKETS TOGETHER

1. We now have the elements we need to understand the movements of output, the interest rate, and the exchange rate.
2. Goods market equilibrium implies that output depends, among other factors, on the interest rate and the exchange rate:

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, E) \quad (8)$$

3. Now, treat the interest rate, i as the policy rate set by the central bank:

$$i = \bar{i} \quad (9)$$

4. And the interest parity condition implies a positive relation between the domestic interest rate and the exchange rate:

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e \quad (10)$$

5. Together, these three relations determine output, the interest rate and the exchange rate.
6. Working with three relations is not very easy, but we can easily reduce them to two by using the interest parity condition to eliminate the exchange rate in the goods market equilibrium relation.
7. Doing this gives us the following two equations, the open-economy versions of our familiar IS and LM relations:

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \frac{1 + i}{1 + i^*} \bar{E}^e) \quad (11)$$

$$i = \bar{i} \quad (12)$$

8. Together, the two equations determine the interest rate and equilibrium output.
9. Then, using Equation (10) we get the implied exchange rate.
10. Take the IS relation first and consider the effects of an increase in the interest rate on output. An increase in the interest rate now has two effects:
 - (a) The first effect, which was already present in a closed economy, is the direct effect on investment: a higher interest rate leads to a decrease in investment, a decrease in the demand for domestic goods and a decrease in output.
 - (b) The second effect, which is present only in the open economy, is the effect through the exchange rate: an increase in the domestic interest rate leads to an increase in the exchange

rate – an appreciation. The appreciation, which makes domestic goods more expensive relative to foreign goods, leads to a decrease in net exports and, therefore, to a decrease in the demand for domestic goods and a decrease in output.

11. Both effects work in the same direction: an increase in the interest rate decreases demand directly and indirectly – through the adverse effect of the appreciation on demand.
12. As an increase in the interest rate reduces output, the IS curve is downward-sloping.
13. The LM curve is the same as in the closed economy: it is a horizontal line, at the level of the interest rate $\bar{i}$ set by the central bank.

[Figure 2: The IS-LM model in an open economy]

14. Equilibrium in the goods and financial markets is attained at point 0 with output level $Y = Y_0$ and interest rate $i = i_0 = \bar{i}$.
15. The equilibrium value of the exchange rate is associated with the equilibrium interest rate, $i = i_0 = \bar{i}$, is equal to $E = E_0 = \bar{E}^e$.
16. In summary,
 - (a) The IS curve is downward-sloping – An increase in the interest rate leads directly and indirectly (through the exchange rate) to a decrease in demand and a decrease in output.
 - (b) The LM curve is horizontal at the interest rate set by the central bank.
17. Equilibrium output and the equilibrium interest rate are given by the intersection of the IS and the LM curves. Given the foreign interest rate and the expected future exchange rate, the equilibrium interest rate determines the equilibrium exchange rate.

THE EFFECTS OF POLICY IN AN OPEN ECONOMY

1. Having derived the IS–LM model for the open economy, we now put it to use and look at the effects of policy.

The Effects of Monetary Policy in an Open Economy

1. At a given level of output, a decrease in the money stock by, say, $\Delta M < 0$ leads to an increase in the interest rate: the LM curve shifts up, from LM_0 to LM_1 . Because money does not directly enter the IS relation, the IS curve does not shift.

[Figure 3: The effect of an increase in the domestic interest rate]

2. The equilibrium moves from point 0 to point 1. The increase in the interest rate leads to an appreciation.
3. So, in an open economy, monetary policy works through two channels:
 - (a) As in the closed economy, it works through the effect of the interest rate on spending.
 - (b) It works through the effect of the interest rate on exchange rate and the effect of the exchange rate on exports and imports.
4. Both effects work in the same direction.
5. A monetary contraction leads to an increase in the interest rate, making domestic bonds more attractive and triggering an appreciation.
6. The higher interest rate and the appreciation both decrease demand and output.

The Effects of Fiscal Policy in an Open Economy with an Unchanged Interest Rate

1. Now that we have looked at monetary policy, we look at an expansionary fiscal policy.
2. Suppose that, starting from a balanced budget, the government decides to increase defence spending without raising taxes and so runs a budget deficit.
3. The economy is initially at point 0 with $Y = Y_0$ and $i = i_0 = \bar{i}$. The increase in government spending by, say, $\Delta G > 0$ increases output at a given interest rate, shifting the IS curve to the right, from IS_0 to IS_1 .
4. Because the central bank does not change the policy rate, the LM curve does not shift.

[Figure 4: The effects of an increase in government with unchanged interest rate]

5. The new equilibrium is at point 1, with a higher level of output at Y_1 .
6. Because the interest rate has not changed, neither has the exchange rate.
7. So, an increase in government spending, when the central bank keeps the interest rate unchanged, leads to an increase in output with no change in exchange rate.
8. What happens to the various components of demand?
 - (a) Consumption and government spending increase: Consumption goes up because of the increase in income and government spending goes up because of the policy.
 - (b) Investment also rises because $I = I(Y, r)$.
 - (c) Since $NX = NX(Y, Y^*, E)$, with Y^* and E unchanged, a higher output leads to an increase in imports and hence a decrease in NX .

The Effects of Fiscal Policy in an Open Economy with the Central Bank Raising Interest Rate

1. Suppose $Y = Y_n$ but government needs to increase spending to pay for an exceptional event, such as a big flood.
2. In this case, the central bank will worry that the increase in G may move the economy above Y_n . As a result, the central bank responds by an increase in the interest rate.
3. In this case, when IS shifts to the right, LM shifts up.

[Figure 5: The effects of an increase in government with higher interest rate]

4. The new equilibrium is at point 1, with a higher level of output and a higher interest rate.
5. The higher interest rate leads to an increase in the exchange rate – an appreciation. So an increase in government spending leads to an increase in output, an increase in the interest rate and an appreciation.
6. Can we tell what happens to the various components of demand?
 - (a) Consumption and government spending both go up. Consumption goes up because $C = C(Y - T)$, and government spending goes up by assumption.
 - (b) What happens to investment is ambiguous. Recall that investment depends on both output and the interest rate: $I = I(Y, i)$. On the one hand, output goes up, leading to an increase in investment. But on the other, the interest rate also goes up, leading to a decrease in investment. Depending on which of these two effects dominates, investment can go up or down.
 - (c) Recall that $NX = NX(Y, Y^*, E)$. Thus, both the increase in output and the appreciation combine to decrease net exports: the increase in output increases imports, and the appreciation decreases exports and increases imports.
7. This version of IS-LM model for an open economy was first put together in the 1960s by Robert Mundell of Columbia and Marcus Fleming of the IMF.

FIXED EXCHANGE RATES

1. We have assumed so far that central bank chose the interest rate and let exchange rate adjust freely.
2. In many countries, central banks may go for different exchange rate regimes.

Pegs, Crawling Pegs, Bands, the EMS and the Euro

1. At one end of the spectrum are countries with **flexible exchange rates**, such as the USA or Japan. These countries do not have explicit exchange rate targets. Although their central banks probably do not ignore movements in the exchange rate, they have shown themselves to be quite willing to let their exchange rates fluctuate considerably.
2. At the other end are countries that operate under **fixed exchange rates**. These countries maintain a fixed exchange rate in terms of some foreign currency. Some peg their currency to the dollar. For example, from 1991–2001, Argentina pegged its currency, the peso, at the highly symbolic exchange rate of 1 dollar for 1 peso. Other countries used to peg their currency to the French franc (most of these are former French colonies in Africa); as the French franc has been replaced by the Euro, they are now pegged to the Euro.
3. Still other countries peg their currency to a basket of foreign currencies, with the weights reflecting the composition of their trade. For example, Singapore.
4. The label ‘fixed’ is a bit misleading: it is not the case that the exchange rate in countries with fixed exchange rates never actually changes. But changes are rare. Because these changes are rare, economists use specific words to distinguish them from the daily changes that occur under flexible exchange rates.
5. A decrease in the exchange rate under a regime of fixed exchange rates is called a **devaluation** rather than a **depreciation**, and an increase in the exchange rate under a regime of fixed exchange rates is called a **revaluation** rather than an **appreciation**.
6. Between these extremes are countries with various degrees of commitment to an exchange rate target. For example, some countries operate under a **crawling peg**.
7. The name describes it well: these countries typically have inflation rates that exceed the US inflation rate. If they were to peg their nominal exchange rate against the dollar, the more rapid increase in their domestic price level above the US price level would lead to a steady real appreciation and rapidly make their goods uncompetitive. To avoid this effect, these countries choose a predetermined rate of depreciation against the dollar. They choose to ‘crawl’ (move slowly) vis-à-vis the dollar.

8. Yet another arrangement is for a group of countries to maintain their bilateral exchange rates (the exchange rate between each pair of countries) within some bands. Perhaps the most prominent example was the European Monetary System (EMS), which determined the movements of exchange rates within the EU from 1978–1998. Under the EMS rules, member countries agreed to maintain their exchange rate vis-à-vis the other currencies in the system within narrow limits, or bands, around a central parity – a given value for the exchange rate. Changes in the central parity and devaluations or revaluations of specific currencies could occur, but only by common agreement among member countries.
9. After a major crisis in 1992, which led a number of countries to drop out of the EMS altogether, exchange rate adjustments became more and more infrequent, leading a number of countries to move one step further and adopt a common currency, the euro. The conversion from domestic currencies to the euro began on 1 January 1999, and was completed in early 2002.

Monetary Policy When the Exchange Rate is Fixed

1. Suppose a country decides to peg its exchange rate at some chosen value, call it $\bar{E}$.
2. How does it actually achieve this? The government cannot just announce the value of the exchange rate and stand there. Rather, it must take measures so that its chosen exchange rate will prevail in the foreign exchange market. Let's look at the implications and mechanics of pegging.
3. Pegging or no pegging, the exchange rate and the nominal interest rate must satisfy the interest parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (13)$$

4. Now suppose a country pegs the exchange rate at $E_t = \bar{E}$, so the current exchange rate $E_t = \bar{E}$. If financial and foreign exchange markets believe that the exchange rate will remain pegged at this value, then their expectation of the future exchange rate, E_{t+1}^e , is also equal to $\bar{E}$, and the interest parity relation becomes:

$$(1 + i_t) = (1 + i_t^*) \quad (14)$$

This means that:

$$i_t = i_t^* \quad (15)$$

5. In words: if financial investors expect the exchange rate to remain unchanged, they will require the same nominal interest rate in both countries. Under a fixed exchange rate and perfect capital mobility, the domestic interest rate must be equal to the foreign interest rate.
6. Under fixed exchange rates, the central bank gives up monetary policy as a policy instrument. With a fixed exchange rate, the domestic interest rate must be equal to the foreign interest rate.

Fiscal Policy When the Exchange Rate is Fixed

1. If monetary policy can no longer be used under fixed exchange rates, what about fiscal policy?
2. The effects of an increase in government spending when the central bank pegs the exchange rate are identical to those for the case of flexible exchange rates and unchanged monetary policy.
3. This is because, under flexible exchange rates, if the increase in spending, is not accompanied by a change in the interest rate, the exchange rate does not move.
4. Thus, when government spending increases, it makes no difference whether the country pegs its exchange rate.
5. The difference between fixed and flexible exchange rate is the ability of the central bank to respond.
6. If the increase in government spending pushed the economy above potential output, thus raising the possibility that inflation might increase, the central bank could respond by raising the interest rate. This option is no longer available under fixed exchange rates because the interest rate must be equal to the foreign rate.
7. Then, why would a country choose to fix its exchange rate? You have seen a number of reasons why this appears to be a bad idea:
 - (a) By fixing the exchange rate, a country gives up a powerful tool for correcting trade imbalances or changing the level of economic activity.
 - (b) By committing to a particular exchange rate, a country also gives up control of its interest rate. Not only that, but the country must match movements in the foreign interest rate, at the risk of unwanted effects on its own activity.
 - (c) Although the country retains control of fiscal policy, one policy instrument may not be enough. For example, a fiscal expansion can help the economy get out of a recession, but only at the cost of a larger trade deficit. And a country that wants, for example, to decrease its budget deficit cannot, under fixed exchange rates, use monetary policy to offset the contractionary effect of its fiscal policy on output.
8. Why, then, do some countries fix their exchange rate?
9. Why have European countries adopted a common currency?